

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Summer Examination – 2025

Course: B.Tech.

Branch : Computer Engineering

Subject Code & Name: BTCOE404B_Y19 Object-Oriented Programming in C++

Max Marks: 60

Date: 09/07/25

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Objective type questions. (Compulsory Question)		12
1 How to access data members of a class?	L2	1
a) Dot, arrow or direct call b) Dot operator		
c) Arrow operator d) Dot or arrow as required		
2 To convert from a user-defined class to a basic type, you would most likely use	L1	1
a) a built-in conversion operator b) a one-argument constructor.		
c) an overloaded = operator d) a conversion operator that's a member of the class.		
3 A pure virtual function is a virtual function that	L1	1
a) causes its class to be abstract. b) returns nothing.		
c) is used in a base class. d) takes no arguments.		
4 Which feature of OOP indicates code reusability?	L2	1
a) Abstraction b) Polymorphism c) Encapsulation d) Inheritance		
5 How many types of access specifiers are provided in OOP (C++)?	L1	1
a) 4 b) 3 c) 2 d) 1		
6 Which among the following represents correct constructor?	L1	1
a) -classname() b) classname() c) ()classname d) ~classname()		
7 Which type of members can't be accessed in derived classes of a base class?	L1	1
a) All can be accessed b) Protected c) Private d) Public		
8 How many arguments are required in the definition of an overloaded	L1	1

	binary operator member function?		
	a) 4 b) 3 c) 2 d) 1		
9	How many instances of an abstract class can be created?	L1	1
	a) 1 b) 5 c) 2 d) 0		
10	Which of the following concepts is used to implement late binding?	L1	1
	a) Virtual function b) Operator function		
	c) Const function d) Static function		
11	To open a file for both reading and writing, which class should be used?	L1	1
	a) ifstream b) ofstream c) fstream d) filebuf		
12	Which of the following is NOT a valid file opening mode in C++?	L1	1
	a) ios::binary b) ios::ate c) ios::start d) ios::noreplace		
Q. 2	Solve the following.		12
	A) Explain the characteristics of object oriented language in detail.	(L2)	6
	B) Define a constructor. Explain the types of constructor and its characteristics.	(L1 & L2)	6
Q.3	Solve the following.		12
	A) Define inheritance. Explain types of inheritance in detail.	(L1 & L2)	6
	B) Write a C++ program to overload binary + operator using member function for addition of two complex numbers.	(L3)	6
Q. 4	Solve Any Two of the following.		12
	A) What are the rules for virtual function? How run time polymorphism is achieved using virtual functions, explain with the help of a C++ program.	(L1 & L3)	6
	B) Explain the following term:	(L2)	6
	i) Pure virtual function		
	ii) this pointer		
	C) What is friend function? Write the difference between member function and friend function.	(L1 & L4)	6

Q.5	Solve Any Two of the following.		12
A)	Explain the manipulators with suitable example.	(L2)	6
B)	Explain the file opening modes in C++.	(L2)	6
C)	Write a C++ program for writing a class object to a file and reading an object from a file.	(L3)	6
Q. 6	Solve Any Two of the following.		12
A)	Define template. Write a C++ program to overload a template function.	(L1 & L3)	6
B)	Explain C++ exception handling mechanism with suitable example.	(L2)	6
C)	Explain the components of STL in short.	(L2)	6
*** End ***			